

System Overview: Nuevo Pipeline Architecture

1 System Overview

This diagram illustrates the detailed architecture of the **nuevo** pipeline, showing internal processing steps and data flows for both main components.

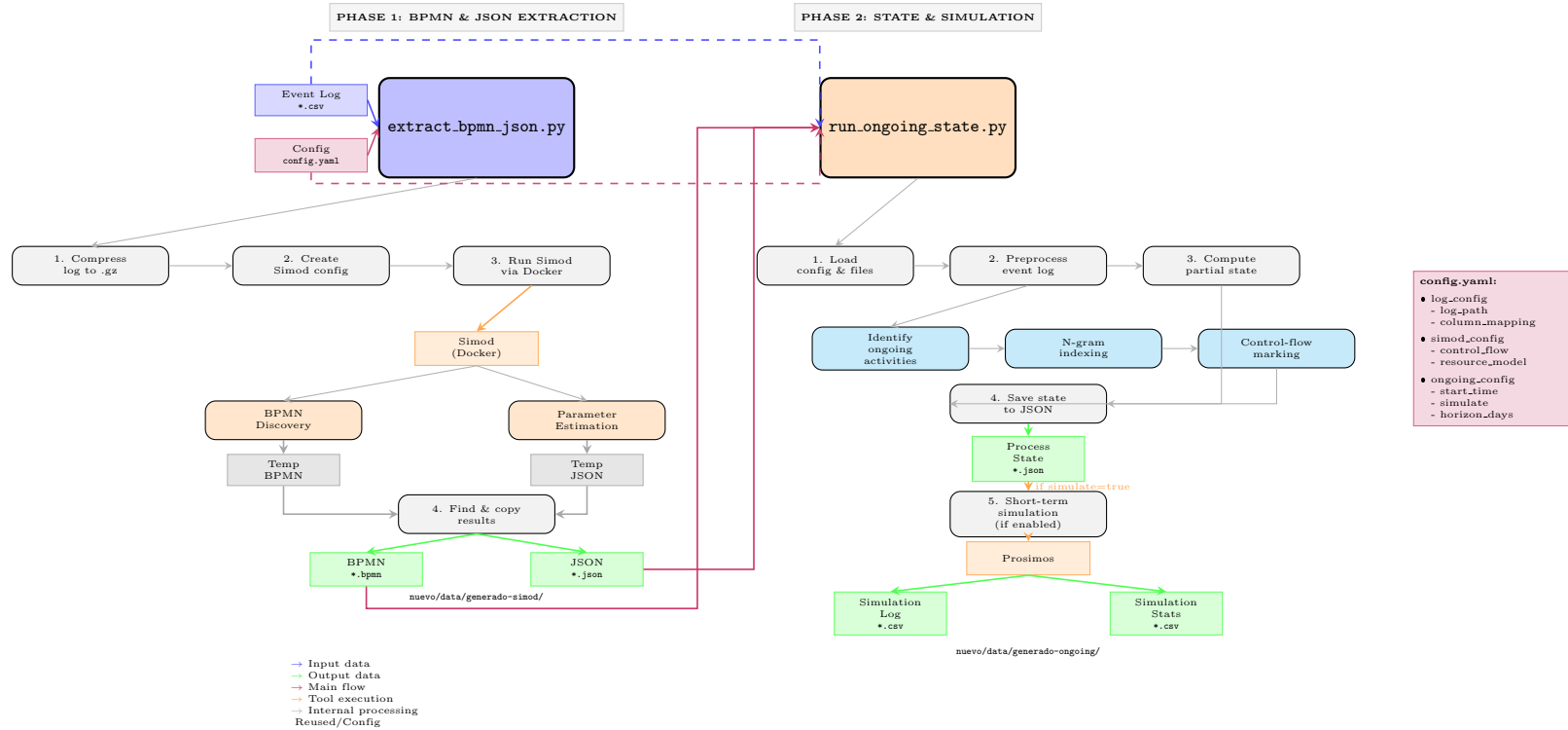


Figure 1: Detailed architecture of the **nuevo** pipeline. **Phase 1** (left): `extract_bpmn_json.py` compresses the event log, creates Simod configuration, executes Simod via Docker for BPMN discovery and parameter estimation, then copies results to `nuevo/data/generado-simod/`. **Phase 2** (right): `run_ongoing_state.py` loads configuration and files, preprocesses the event log, computes partial process state (identifying ongoing activities, building N-gram indices, computing control-flow markings), and optionally runs short-term simulation via Prosimos, producing outputs in `nuevo/data/generado-ongoing/`.

1.1 Phase 1: BPMN & JSON Extraction Details

The `extract_bpmn_json.py` script performs the following steps:

1. **Log Compression:** Compresses the input CSV event log to `.gz` format for Simod processing
2. **Configuration Creation:** Generates a Simod-specific `configuration.yaml` file from `config.yaml`, including:
 - Log column mappings (case, activity, resource, start_time, end_time)
 - Control-flow discovery parameters (mining algorithm, epsilon, eta)
 - Resource model settings (profiles, granularity, confidence)
 - Extraneous activity delays configuration
3. **Simod Execution:** Runs Simod via Docker container (`nokal/simod`) which:
 - Discovers the BPMN process model structure
 - Estimates stochastic parameters (arrival distributions, resource profiles, task durations, gateway probabilities)
4. **Result Extraction:** Locates the most recent timestamped output directory, finds `best_result` subdirectory, and copies `*.bpmn` and `*.json` files to `data/generado-simod/`

1.2 Phase 2: State & Simulation Details

The `run_ongoing_state.py` script performs the following steps:

1. **Configuration & File Loading:**
 - Loads `config.yaml` for ongoing-state settings
 - Locates BPMN and JSON files (first in `data/generado-simod/`, then in base directory)
 - Loads the original event log with column mapping
2. **Event Log Preprocessing:**
 - Applies column mapping to standardize column names
 - Filters events by cut-off time (`start_time ≤ cut-off`)
 - Identifies ongoing cases (`end_time` is null or after cut-off)
 - Normalizes timestamps to UTC
3. **Partial State Computation:**
 - **Ongoing Activities:** Identifies activities currently running (no `end_time` or `end_time` after cut-off)
 - **N-gram Indexing:** Builds extended BPMN model (tasks split into +START/+COMPLETE), computes reachability graph, and creates N-gram index for state lookup
 - **Control-flow Marking:** For each case, serializes event history, queries N-gram index for marking, and determines token distribution on flows and activities

- **Enabled Elements:** Computes enabled activities, gateways, and events based on markings and ongoing activities
 - **State Serialization:** Generates a JSON structure containing `last_case_arrival` and per-case states with control-flow state, ongoing activities, and enabled elements
4. **Save State to JSON:** Writes the computed partial state to `output.json` (later renamed to `{log_name}_process_state.json`) in the output directory
5. **Short-term Simulation** (if `simulate: true`):
- Calculates simulation horizon (from `horizon_days` or explicit `simulation_horizon`)
 - Executes Prosimos with partial state as initial condition
 - Generates simulation log and statistics CSV files

1.3 Data Structures

- **Process State JSON:** Contains `last_case_arrival` and per-case states with:
 - `control_flow_state`: Token distribution on flows and activities
 - `ongoing_activities`: List of running activities with resources and timestamps
 - `enabled_activities`: Activities ready to start with enabled times
 - `enabled_gateways`: Gateways ready to fire
 - `enabled_events`: Events ready to occur
- **Simulation Log CSV:** Event log format with case ID, activity, start time, end time, resource
- **Simulation Stats CSV:** Aggregated KPIs (cycle time, waiting time, resource utilization, etc.)